BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

October 21, 2026

Question 1

Question 2

2a)

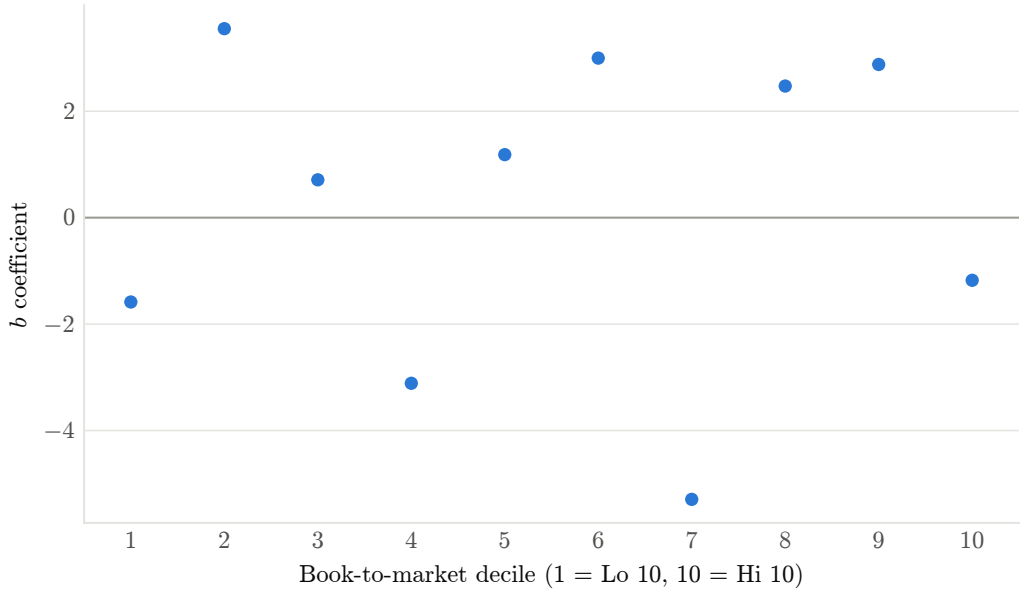


Figure 1: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted book-to-market decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2b)

The sample (Pearson) correlation between the monthly SDF series $M^*_{\text{BM},t}$ estimated in 2a) and $M^*_{\text{MOM},t}$ estimated in 2b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\text{Corr}}\left(M^*_{\text{BM},t},\, M^*_{\text{MOM},t}\right) = 0.355.$$

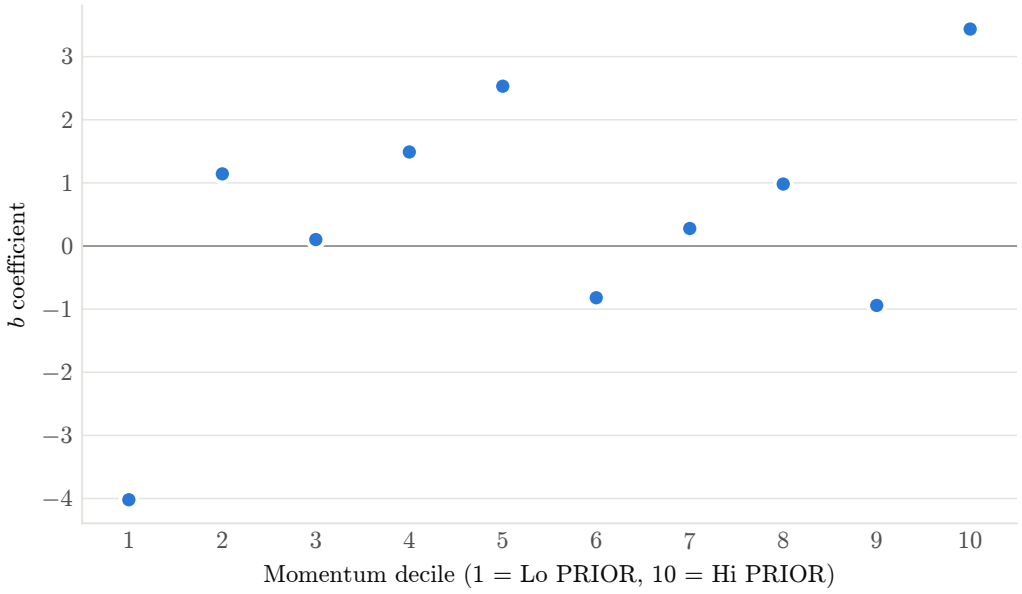


Figure 2: Coefficients b of the SDF $M_t^* = a - b' x R_t$ that prices the monthly excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2c)

Table 1: Monthly Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = b' x R_t$ implied by the SDFs of 2a)–2c), July 1927 to August 2026 ($T = 1190$ months). Not annualized.

SDF	Test assets	$\widehat{SR}(xR^*)$
2a)	10 book-to-market deciles	0.192
2b)	10 momentum deciles	0.258
2c)	20 book-to-market and momentum deciles	0.305

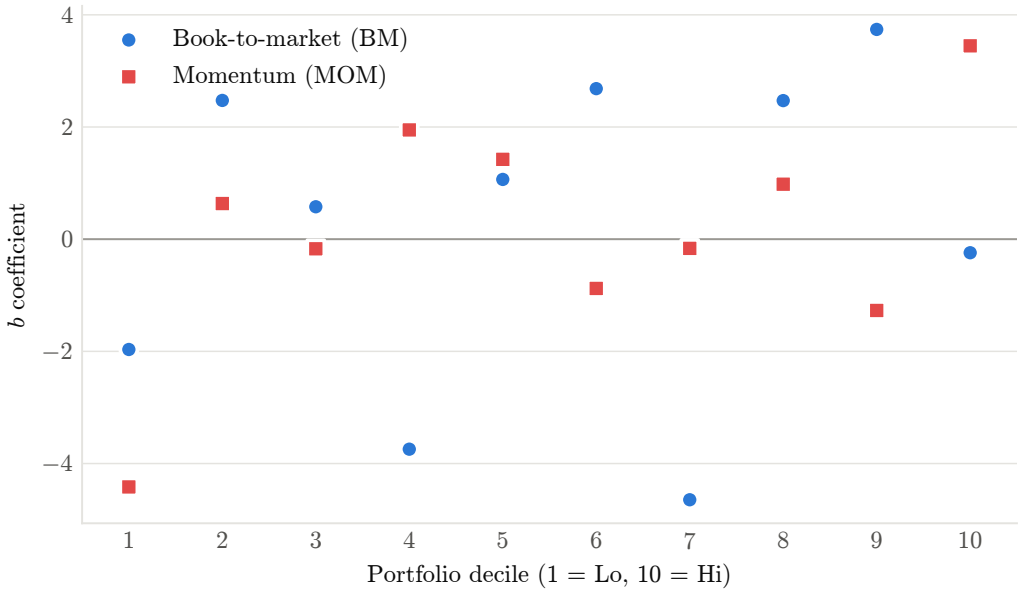


Figure 3: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 20 value-weighted portfolios in the merge of the book-to-market (blue circles) and momentum (red squares) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2d)

[Explanation will be here]

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in 2a) (10 book-to-market deciles), 2b) (10 momentum deciles) and 2c) (all 20 portfolios). Monthly excess returns in decimal units, July 1927 to August 2026 ($T = 1190$ months); covariances divide by T.

	SDF M_t^* estimated in		
Portfolio	2a) BM	2b) MOM	2c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.0000	−0.0015	0.0000
2	0.0000	−0.0001	0.0000
3	0.0000	−0.0002	0.0000
4	0.0000	−0.0008	0.0000
5	0.0000	0.0005	0.0000
6	0.0000	0.0013	0.0000
7	0.0000	0.0003	0.0000
8	0.0000	0.0024	0.0000
9	0.0000	0.0034	0.0000
10 (Hi 10)	0.0000	0.0032	0.0000
Panel B: momentum deciles			
1 (Lo PRIOR)	−0.0102	0.0000	0.0000
2	−0.0051	0.0000	0.0000
3	−0.0033	0.0000	0.0000
4	−0.0015	0.0000	0.0000
5	−0.0009	0.0000	0.0000
6	−0.0003	0.0000	0.0000
7	0.0005	0.0000	0.0000
8	0.0019	0.0000	0.0000
9	0.0022	0.0000	0.0000
10 (Hi PRIOR)	0.0054	0.0000	0.0000

Question 3

Question 4